Chapter 4 • Factorization and Algebraic Manipulation

# Solved Exercise 4.2

**1. Factorize by writing each quartic as $A^2-B^2$.**
(i) $4x^4+81y^4 = (2x^2+9y^2)^2-(6xy)^2$
$= (2x^2-6xy+9y^2)(2x^2+6xy+9y^2)$.
(ii) $a^4+64b^4 = (a^2+8b^2)^2-(4ab)^2$
$= (a^2-4ab+8b^2)(a^2+4ab+8b^2)$.
(iii) $x^4+4x^2+16 = (x^2+4)^2-(2x)^2$
$= (x^2-2x+4)(x^2+2x+4)$.
(iv) $x^4-14x^2+1 = (x^2+1)^2-(4x)^2$
$= (x^2-4x+1)(x^2+4x+1)$.
(v) $x^4-30x^2y^2+9y^4 = (x^2+3y^2)^2-(6xy)^2$
$= (x^2-6xy+3y^2)(x^2+6xy+3y^2)$.
(vi) $x^4-7x^2y^2+y^4 = (x^2+y^2)^2-(3xy)^2$
$= (x^2-3xy+y^2)(x^2+3xy+y^2)$.

**2(i). Factorize $(x+1)(x+2)(x+3)(x+4)+1$.**
Pair $(x+1)(x+4)=x^2+5x+4$ and $(x+2)(x+3)=x^2+5x+6$.
Let $u=x^2+5x+5$. Then $(u-1)(u+1)+1=u^2$.
**$(x^2+5x+5)^2$.**

**2(ii). Factorize $(x+2)(x-7)(x-4)(x-1)+17$.**
Products are $x^2-5x-14$ and $x^2-5x+4$. Let $u=x^2-5x-5$.
$(u-9)(u+9)+17 = u^2-64$
$= (u-8)(u+8)$.
**$(x^2-5x-13)(x^2-5x+3)$.**

**2(iii). Factorize $(2x^2+7x+3)(2x^2+7x+5)+1$.**
Let $u=2x^2+7x+4$. Then $(u-1)(u+1)+1=u^2$.
**$(2x^2+7x+4)^2$.**

**2(iv). Factorize $(3x^2+5x+3)(3x^2+5x+5)-3$.**
Let $u=3x^2+5x+4$. Then $(u-1)(u+1)-3=u^2-4=(u-2)(u+2)$.
**$(3x^2+5x+2)(3x^2+5x+6)$.**

**2(v). Factorize $(x+1)(x+2)(x+3)(x+6)-3x^2$.**
Expansion gives $x^4+12x^3+44x^2+72x+36$.
Compare with $(x^2+4x+6)(x^2+8x+6)$; expansion gives exactly the same polynomial.
**$(x^2+4x+6)(x^2+8x+6)$.**

**2(vi). Factorize $(x+1)(x-1)(x+2)(x-2)-16x^2$.**
$(x^2-1)(x^2-4)-16x^2 = x^4-21x^2+4$
$= (x^2+2)^2-(5x)^2$.
**$(x^2-5x+2)(x^2+5x+2)$.**

**3. Factorize using perfect-cube expansions.**
(i) $8x^3+12x^2+6x+1 = (2x)^3+3(2x)^2(1)+3(2x)(1^2)+1^3$
$= (2x+1)^3$.
(ii) $27a^3+108a^2b+144ab^2+64b^3 = (3a)^3+3(3a)^2(4b)+3(3a)(4b)^2+(4b)^3$
$= (3a+4b)^3$.
(iii) $x^3+18x^2y+108xy^2+216y^3 = x^3+3x^2(6y)+3x(6y)^2+(6y)^3$
$= (x+6y)^3$.
(iv) $8x^3-60x^2y+150xy^2-125y^3 = (2x)^3-3(2x)^2(5y)+3(2x)(5y)^2-(5y)^3$
$= (2x-5y)^3$.

**4. Factorize using $a^3\pm b^3=(a\pm b)(a^2\mp ab+b^2)$.**
(i) $125a^3-1=(5a-1)(25a^2+5a+1)$.
(ii) $64x^3+125=(4x+5)(16x^2-20x+25)$.
(iii) $x^6-27=(x^2-3)(x^4+3x^2+9)$.
(iv) $1000a^3+1=(10a+1)(100a^2-10a+1)$.
(v) $343x^3+216=(7x+6)(49x^2-42x+36)$.
(vi) $27-512y^3=(3-8y)(9+24y+64y^2)$.

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